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WORKSHEET

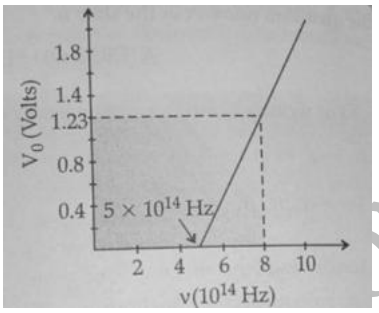
CLASS XII

PHYSICS

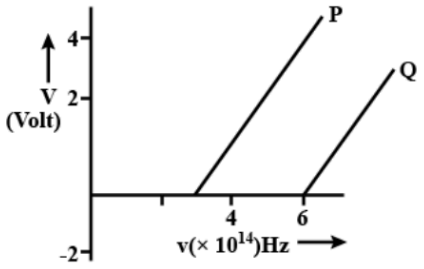
DUAL NATURE OF RADIATION AND MATTER

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1. A monochromatic source, emitting light of wavelength, 600 nm, has a power output of 66 W. Calculate the number of photons emitted by this source in 2 minutes.
2. Using the graph shown in the figure. For stopping potential vs. the incident frequency of photons, calculate Planck's constant.



3. In the study of photoelectric effect, the graph between the stopping potential V and frequency υ of the incident radiation on two different metals P and Q is shown below:
   (i) Which one of the two metals has higher threshold frequency?
   (ii) Determine the work function of the metal which has greater value.
   (iii) Find the maximum kinetic energy of electron emitted by the light of frequency 8 x 10^{14} Hz for this metal.



4. By how much would the stopping potential for a given photosensitive surface go up if the frequency of incident radiations were to be increased from 4 x 10^{15} Hz to 8 x 10^{15} Hz? Given, h= 6.4 x 10^{-34} Js, e= 1.6 x 10^{-19} C and c= 3 x 10^{8} m/s.
5. Calculate the threshold frequency of photons for photoelectric emission from a metal of work function 0.1 eV.
6. The following table gives the values of work function for a few photosensitive metals:

S. No.	Metal	Work Function (eV)
1.	Na	1.92
2.	K	2.15
3.	Mo	4.17

If each of these metals is exposed to radiations of wavelength 300 nm, which of them will not emit photoelectrons and why?

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7. Find the frequency of light which ejects electrons from a metal surface, fully stopped by a retarding potential of 3.3 V. If photo electric emission begins in this metal at a frequency of 8×10^{14} Hz, calculate the work function (in eV) for this metal.
8. The equivalent wavelength of a moving electron has the same value as that of a photon having an energy of 6×10^{-17} J. Calculate the momentum of the electron.
9. Draw a graph between the frequency of incident radiation (υ) and the maximum kinetic energy of the electrons emitted from the surface of a photosensitive material. State clearly how this graph can be used to determine (i) Planck's constant and (ii) work function of the metal.
10. A proton and an alpha particle are accelerated through the same potential. Which one of the two has (i) greater value of de Broglie wavelength associated with it, and (ii) less kinetic energy? Justify your answer.

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